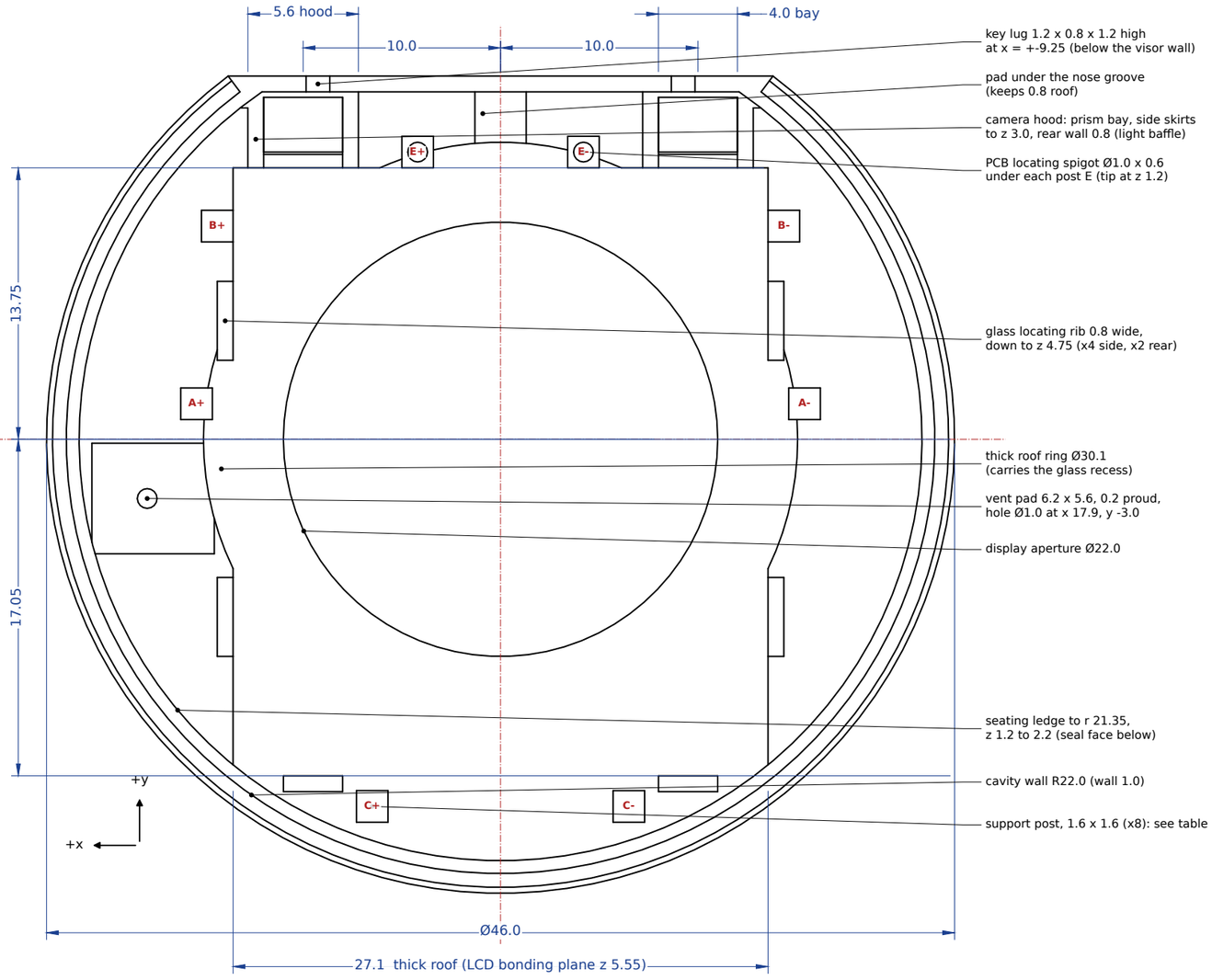
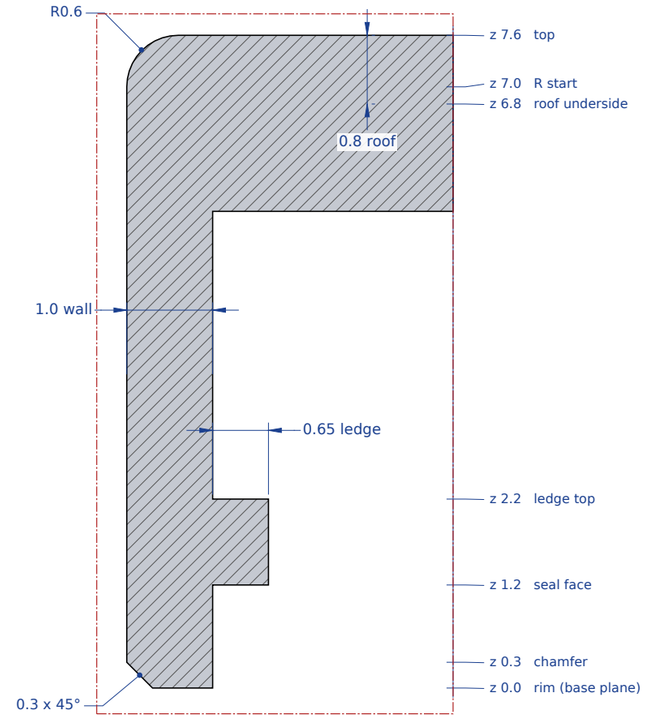
VIEW FROM BELOW - into the cavity 3.5:1 (+x is to the LEFT, visor at the top)



DETAIL E - wall, ledge and roof at the rear (section x = 0) 14:1



Post	x centre	y centre	from - to
A-	-15.4	1.8	PCB top z 1.8 - roof
A+	15.4	1.8	PCB top z 1.8 - roof
B-	-14.35	10.8	PCB top z 1.8 - roof
B+	14.35	10.8	PCB top z 1.8 - roof
C-	-6.5	-18.6	PCB top z 1.8 - roof
C+	6.5	-18.6	PCB top z 1.8 - roof
E-	-4.2	14.55	PCB top z 1.8 - roof
E+	4.2	14.55	PCB top z 1.8 - roof

Posts 1.6 x 1.6 mm = YAML tub_keepouts (contract with the PCB layout, proven in check 12).

NOTES

1. Material HP 3D High Reusability PA 12, Multi Jet Fusion, dyed black, chemical vapor smoothing. Print roof-down (cosmetic top on the bed side, cavity open upward).
2. Minimum wall 0.8 mm by design. Recorded exceptions: the 0.525 mm strip behind each window recess (D04, accepted), the feet of chamfered edges (side wall 0.7 at the rim, hood rear wall 0.681) and the radiused land beside each recess. See export/check_report.md, check 5.
3. No closed void and no cavity without a 2 mm escape path (check 6). Depowder the two camera bays, the visor openings and the Ø1.0 vent hole with care.
4. Critical-to-function: bonding plane z 5.55 (LCD), recess floor z 6.35 (cover glass), seal face z 1.2, post feet z 1.8 (all in one plane +-0.1), window recesses. First-article inspection of these is a release gate. Nothing has been built.
5. Drop-in fits are 0.25 mm nominal. The two spigots (Ø1.0 in a Ø1.15 PCB hole and a 1.15 x 1.65 slot) locate the PCB: gauge them at first article (D19).
6. Camera hoods stay inside the YAML keep-out zones (camera x +-2.85, y 13.7 to 17.3); skirts end at z 3.0 (1.2 above the PCB).

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

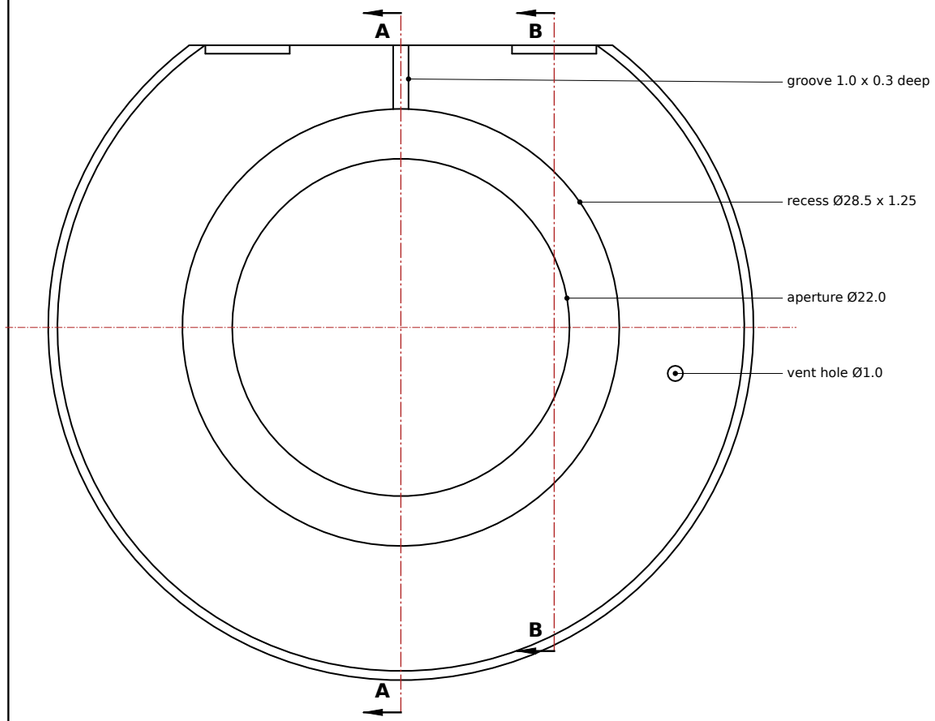
TITLE
GL1-010 Tub (top shell)

PROJECT / PART
GreenLine GL-1 electronic ball marker - tub

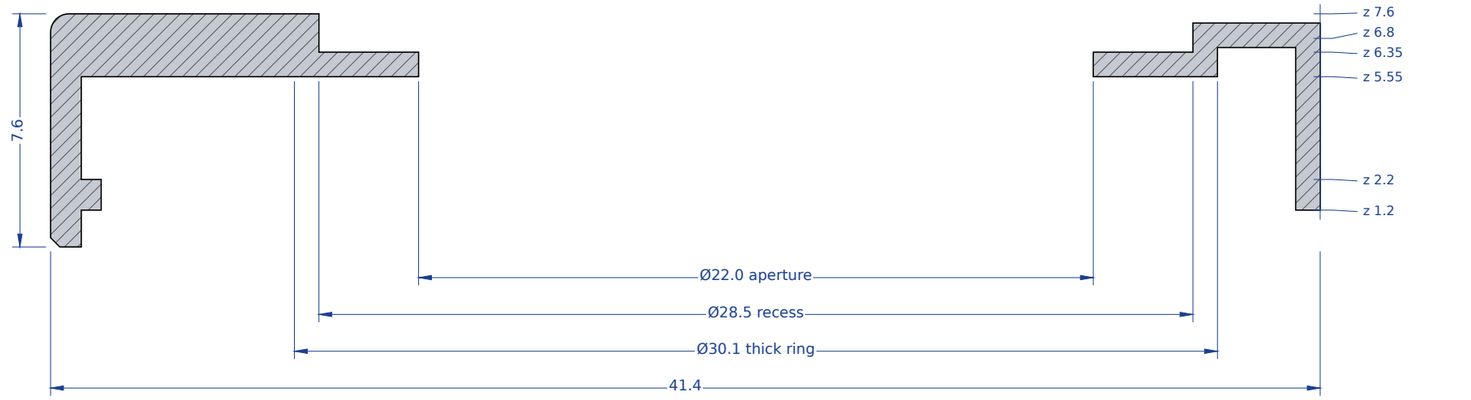
MATERIAL / FINISH	PROCESS
PA12 (HP 3D HR PA 12), dyed black	HP Multi Jet Fusion + vapor smoothing
GENERAL TOLERANCE	SCALE (A3) / UNITS
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	3.5:1 / 14:1 mm

STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	C-EVT1 / C-EVT1	1 of 2 2026-11-11

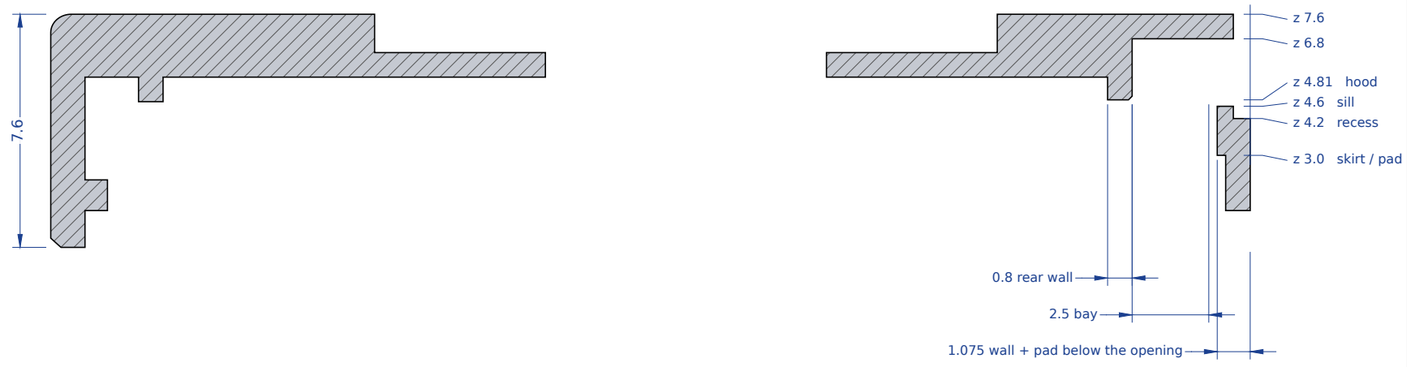
TOP VIEW 2.5:1 (section planes A-A and B-B)



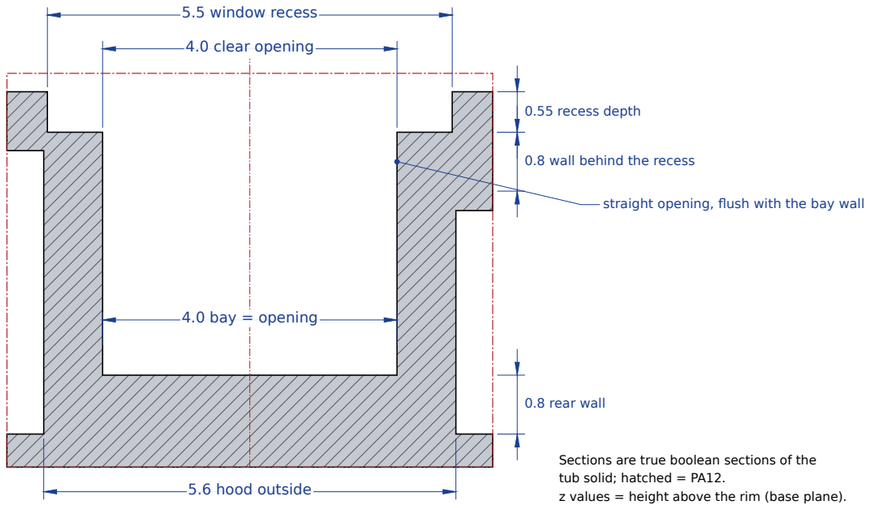
SECTION A-A (x = 0) 5:1



SECTION B-B (x = 10.0, through the camera R hood) 5:1

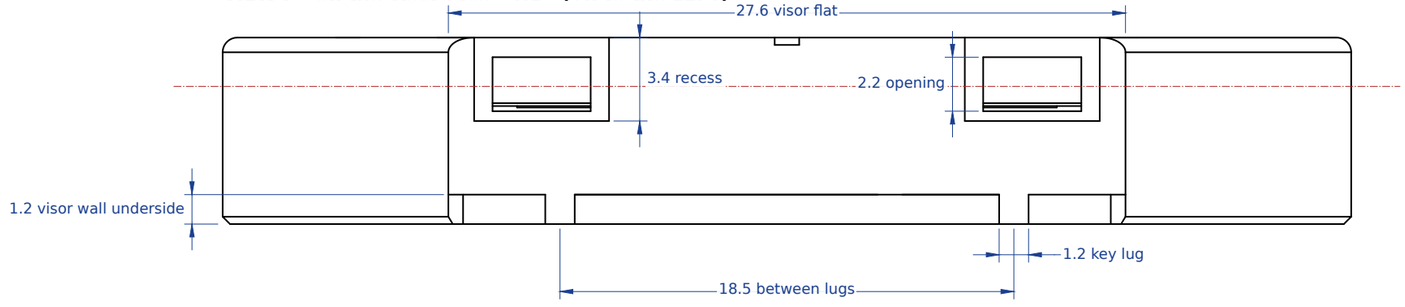


DETAIL G - camera R opening, section at z = 5.61 (seen from above) 12:1



Sections are true boolean sections of the tub solid; hatched = PA12.
z values = height above the rim (base plane).

VIEW F - on the visor face 4:1 (+x to the LEFT)



TITLE	**GL1-010 Tub (top shell)**
PROJECT / PART	GreenLine GL-1 electronic ball marker - tub
MATERIAL / FINISH	PA12 (HP 3D HR PA 12), dyed black
PROCESS	HP Multi Jet Fusion + vapor smoothing
GENERAL TOLERANCE	printed: +-0.3 or +-0.3 % (bureau MJF), as printed
SCALE (A3) / UNITS	2.5:1 / 5:1 / 12:1 mm
STATUS	**PROVISIONAL - not released**
DWG REV / PARAMS	C-EVT1 / C-EVT1
SHEET / DATE	2 of 2 2026-11-11